

Title: “How S&P 500 Index be Predicted Using Consumer Price Index, Unemployment Rate, Foreign Direct Investments & GDP Growth Rate of United States”

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Introduction:

Understanding how financial markets interact with macroeconomic factors is crucial for analyzing economic trends, forecasting market behaviors and making well informed investment decisions. This work contributes to our academic objectives by integrating financial theory with empirical data analysis, enabling us to use econometric methods to fully understand S&P 500 Index. By evaluating how GDP growth rate, Consumer Price Index (CPI), unemployment rate, and foreign direct investments affect the S&P 500 Index, it will not only answer a crucial question about economy but also provide us important data analysis techniques.

Relevant Literature:

The research article – “*Statistically Significant Relationships between Returns on FTSE 100, S&P 500 Market Indexes and Macroeconomic Variables with Emphasis on Unconventional Monetary Policy*” highlights the complex interplay between macroeconomic indicators and the S&P 500 Index. While factors like inflation, unemployment, foreign investment, and GDP growth influence market behavior, their predictive power varies.

Key findings include:

- Inflation: Significant long-term impact on stock returns.
- Unemployment: Critical signal for economic health and market trends.
- Foreign Investment: Indicator of broader economic confidence.
- GDP Growth: Fundamental measure of economic performance and market valuation.

To fully understand these relationships, advanced statistical techniques like vector error correction models and cointegration analysis are essential. The research underscores the need for a comprehensive, multivariate approach rather than relying on a single indicator.

The research article – “*Drivers of S&P 500’s Profitability: Implications for Investment Strategy and Risk Management*” highlights the analysis of S&P 500 Index prediction through macroeconomic variables reveals a complex, dynamic relationship between economic indicators and stock market performance. Key findings highlight that:

- Consumer Price Index and unemployment rates significantly influence market returns
- Foreign Direct Investments and GDP growth provide critical predictive insights
- Non-linear, multifaceted interactions require sophisticated analytical approaches

The research underscores the importance of comprehensive, integrated economic analysis for effective investment strategies. Future research should focus on advanced predictive modeling, incorporating technological indicators and developing more nuanced risk management frameworks. Ultimately,

while no single variable guarantees precise market prediction, a holistic understanding of macroeconomic dynamics can enhance investment decision-making and market comprehension.

The research article – “*An Analysis of Several Factors Affecting the U.S. Dollar Index*” highlights the prediction of S&P 500 Index through macroeconomic variables reveals complex interdependencies:

Key Insights:

- Consumer Price Index (CPI) serves as a critical inflation indicator
- Unemployment rates reflect economic productivity and market sentiment
- Foreign Direct Investments signal international economic confidence
- GDP growth rate represents fundamental economic performance

Predictive Modeling Considerations:

- Multivariate approach essential for comprehensive analysis
- Non-linear relationships require sophisticated econometric techniques
- No single variable provides definitive market prediction

Strategic Implications:

- Holistic economic analysis enhances investment decision-making
- Understanding macroeconomic interactions crucial for risk management
- Continuous model refinement needed due to dynamic market conditions

Future Research Recommendations:

- Advanced predictive modeling techniques
- Integration of emerging economic indicators
- Exploring complex interrelationships between variables

The S&P 500 Index prediction underscores the complexity of financial markets, demanding nuanced, interdisciplinary analytical frameworks.

Research Hypothesis:

The primary hypothesis of this research is that macroeconomic factors significantly influence the Standard & Poor's 500 (S&P 500) Index. Specifically, it is hypothesized that:

1. A higher GDP growth rate will positively impact the S&P 500 Index. As economic expansion tends to enhance corporate profits and investor confidence, leading to an upward trend in stock market performance.
2. Increased foreign direct investment (FDI) will have a positive effect on the S&P 500 Index. This is because FDI often reflects growing investor confidence in the economy, providing additional capital for business expansion and innovation which supports market growth.
3. Rising consumer price index (CPI) within moderate range is hypothesized to positively affect S&P 500 Index as it indicates healthy consumer demand. However, excessive inflation may have an adverse impact by increasing costs for business and reducing purchasing power.

4. Conversely, a higher unemployment rate is expected to negatively affect the S&P 500 Index. As it reflects weak economic conditions, reduced consumer spending and potential decline in corporate earnings.

Empirical Methodology: Data & Sample Construction

$$Y = \beta_0 + (\beta_1 \cdot X_1) + (\beta_2 \cdot X_2) + (\beta_3 \cdot X_3) + (\beta_4 \cdot X_4) + \varepsilon$$

where,

- Y = S&P 500 Index
- X_1 = GDP Growth Rate _(USA)
- X_2 = Consumer Price Index
- X_3 = Unemployment Rate
- X_4 = Foreign Direct Investment
- $\beta_0, \beta_1, \beta_3, \beta_4$ = Coefficient to be estimated
- ε = Error term

In this model, Y represents the Standard & Poor's 500 Index (S&P 500), X_1 signifies the GDP growth rate of United States, X_2 indicates the consumer price index, X_3 talks about the unemployment rate in United States and X_4 signifies the foreign direct investment. The dataset spans a timeline of 34 years, with data recorded on a monthly basis.

In the above regression equation, β_1 indicates the change in the S&P 500 Index for 1 unit change in GDP growth rate, β_2 indicates the change in the S&P 500 Index for 1 unit change in Consumer Price Index (CPI). Similarly, β_3 and β_4 explain the change in S&P 500 Index for 1 unit change in unemployment rate and foreign direct invest respectively. These coefficients will offer insights into quantifying the respective factors and enhancing our understanding of the S&P 500 Index.

Results & Discussion:

The summary statistics show that for each of these variables, there are 383 observations (Observations column). The Mean, Standard Deviation (Std. dev.), Minimum, and Maximum values.

Variable	Obs	Mean	Std. dev.	Min	Max
SPIndex	383	1668.942	1081.305	403.69	4769.83
GDPGrowthRate	383	.0021848	.0083927	-.09051	.054805
Unemployment	383	.0572977	.0179251	.034	.148
FDI	383	210829	160744.9	-293968	956604
CPI	383	208.515	43.63852	138.6	308.742

Table (1.1) : Summary Statistics (Pre Log)

Source	SS	df	MS	Number of obs	=	383
Model	386429205	4	96607301.2	F(4, 378)	=	606.47
Residual	60212878.7	378	159293.33	Prob > F	=	0.0000
				R-squared	=	0.8652
				Adj R-squared	=	0.8638
Total	446642084	382	1169220.11	Root MSE	=	399.12

SPIndex	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
GDPGrowthRate	5647.11	2440.301	2.31	0.021	848.8449	10445.38
CPI	21.91186	.5336894	41.06	0.000	20.86249	22.96123
Unemployment	-15225.64	1172.636	-12.98	0.000	-17531.34	-12919.93
FDI	-.0002009	.0001481	-1.36	0.176	-.0004922	.0000903
_cons	-1997.596	126.2217	-15.83	0.000	-2245.781	-1749.412

Table (1.2) : OLS Regression Results (Pre Log)

The S&P 500 Index has a mean of 1668.942 and a standard deviation of 1081.305, indicating significant volatility in the stock market over the sample period. The GDP Growth Rate has a mean of 0.0211848, or about 2.1% annual growth on average. The Unemployment rate has a mean of 0.0572977, or around 5.7% unemployment. FDI has a highly variable range, with a minimum of -293968 and a maximum of 956604. CPI has a mean of 208.515, indicating general inflation over the period.

Table (1.2) presents the regression results derived from our analysis. In alignment with our research hypothesis, the GDP growth rate positively influences the S&P 500 Index, while the unemployment rate negatively affects it. Additionally, the Consumer Price Index (CPI) has a positive effect on the S&P 500 Index. These relationships underscore the theoretical expectations about the dynamics between macroeconomic variables and stock market performance.

To determine the statistical significance of the independent variables and confirm whether they are different from zero, I relied on p-values, t-statistics, and confidence intervals. For this discussion, the focus was on the interpretation of p-values. A p-value measures the probability of obtaining the observed results under the null hypothesis, which assumes that the coefficient of an independent variable equals zero. If the p-value is less than the predefined confidence level (commonly 5% or 1%), reject the null hypothesis, concluding that the independent variable is statistically significant and its effect is different from zero.

The reason for taking the log of the variables is to address the positive bias and skewness often seen in economic time series data. Variables like stock market indexes, GDP, and inflation tend to have distributions with long right tails, making the relationships between them less linear. Taking the log of these variables helps normalize the distribution and make the analysis more statistically robust.

Variable	Obs	Mean	Std. dev.	Min	Max
Log_SPIndex	383	7.22771	.6239169	6.000647	8.470066
Log_GDP	259	-5.603154	1.039156	-11.16227	-2.903974
Unemployment	383	.0572977	.0179251	.034	.148
Log_FDI	368	11.97957	1.135488	3.332205	13.77114
Log_CPI	383	5.318003	.210881	4.931592	5.732506

Table (2.1) : Summary Statistics (Post Log)

Source	SS	df	MS	Number of obs	=	251
Model	94.0857739	4	23.5214435	F(4, 246)	=	717.93
Residual	8.05969946	246	.032763006	Prob > F	=	0.0000
				R-squared	=	0.9211
				Adj R-squared	=	0.9198
Total	102.145473	250	.408581893	Root MSE	=	.18101

Log_SPIndex	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
Log_GDP	.0381461	.0111316	3.43	0.001	.0162207	.0600715
Unemployment	-9.002192	.6838379	-13.16	0.000	-10.34912	-7.655268
Log_FDI	.0233848	.0105274	2.22	0.027	.0026494	.0441201
Log_CPI	2.607968	.0604615	43.13	0.000	2.488879	2.727056
_cons	-6.175962	.295654	-20.89	0.000	-6.758298	-5.593626

Table (2.2) : OLS Regression Results (Post Log)

$$\text{Log_SPIndex} = -6.175962 + 0.0381461 \text{Log_GDP} - 9.002192 (\text{unemployment}) + 0.0233848 \text{Log_FDI} + 2.607968 \text{Log_CPI} + \varepsilon$$

Log_SPIndex is the log-transformed S&P 500 Index, which we want to predict. The coefficient of 0.0381461 for Log_GDP indicates that a 1% increase in logged GDP is associated with a 0.0381461 (or about 3.81%) increase in the logged S&P 500 Index, holding other factors constant. This suggests GDP growth has a positive relationship with stock market performance.

The logged GDP growth rate has a coefficient of 0.0381461 and a t-statistic of 3.43, which is statistically significant at the 1% level. This implies GDP growth is an important predictor of the S&P 500 Index.

The logged unemployment rate has a coefficient of -9.002192, with a highly significant t-statistic of -13.16. This indicates that a 1% increase in the logged unemployment rate is associated with a 9.002192% decrease in the logged S&P 500 Index. Higher unemployment is thus a strong negative predictor of stock market performance.

The logged foreign direct investment has a coefficient of 0.0233848 and a t-statistic of 2.22, which is significant at the 5% level. This suggests FDI inflows also have a positive relationship with the S&P 500 Index.

The logged consumer price index has a coefficient of 2.607968 and a very high t-statistic of 43.13, indicating inflation is a powerful positive predictor of the stock market index. However, this large coefficient may be due to the positive bias in CPI data, which is why the log transformation is so important.

The regression results in Table (2.2) show that the GDP growth rate (X_1), unemployment rate (X_2), foreign direct investment (X_3) and consumer price index (X_4) are statistically significant at a 5% confidence level, indicating that these variables contribute meaningfully to the model. Furthermore, GDP growth rate (X_1), unemployment rate (X_2), and consumer price index (X_4) are statistically significant at a 1% confidence level reflecting an even stronger level of confidence in their relationship with the S&P 500 Index. So from our data and OLS regression, we can say that our model is statistically and economically significant. Economic significance refers to the practical or real-world importance of the results of a study, particularly in terms of their financial or economic implications. It goes beyond mere statistical significance (which indicates whether an effect exists) and addresses whether the size of the effect or its consequences are meaningful or valuable in economic terms.

The R-squared values indicate the model explains about 92% of the variation in the logged S&P 500 Index. Overall, this analysis suggests that macroeconomic factors like GDP growth, unemployment, FDI, and inflation (as measured by CPI) can be powerful predictors of the S&P 500 Index when properly transformed and modelled.

These findings demonstrate the robustness of our model and provide empirical support for the hypothesized relationships. The GDP growth rate's positive impact highlights its role as a key indicator of economic strength, while the unemployment rate's negative relationship underscores the adverse effects of labour market weakness on investor sentiment and stock market performance. The positive influence of the CPI on the S&P 500 Index may reflect inflationary pressures that often align with economic growth and corporate profitability. Together, these results contribute to a deeper understanding of the factors driving the S&P 500 Index and affirm the relevance of these macroeconomic indicators in explaining stock market movements.

Our regression analysis provides evidence supporting our hypothesis; however, we cannot assert with complete certainty that the model accurately predicts the S&P 500 Index. This uncertainty stems from the substantial likelihood of omitted variable bias, which is a common issue in regression analysis. Omitted variable bias occurs when a key independent variable that significantly influences the dependent variable is excluded from the model. Such exclusions can lead to biased and inconsistent parameter estimates, particularly when the omitted variable is correlated with one or more of the independent variables included in the model.

In the context of our analysis, several critical innate factors may have been excluded, potentially impacting the accuracy and robustness of our model. These factors include investor sentiment, which reflects market psychology and plays a significant role in stock price movements; corporate governance and management quality, which are crucial for assessing the overall health and sustainability of the companies within the index; and changes to the calculation methodology or weight adjustments of the index, which can influence its performance. Incorporating these variables into the model could enhance its reliability and improve its predictive power, allowing for a more comprehensive understanding of the dynamics affecting the S&P 500 Index.

Conclusion:

The S&P 500 Index's impressive performance and the probability of the 500 biggest publicly traded corporations in the US are indicators of the overall robust economic growth. Good S&P 500 performance is indicated by economic growth, company profitability, investor confidence, market liquidity and high employment and pay. Regression study shows that macroeconomic variables such as the GDP growth rate, foreign direct investment (FDI), unemployment rate and consumer price index (CPI) have a considerable impact on the S&P 500 fluctuations. However, it's crucial to remember that the success of the stock market doesn't always match that of the actual economy. Independent of the underlying state of the economy, variables including the market mood, geopolitical developments or outside shocks, TO enhance forecast accuracy, future research should examine nonlinear correlations or include more variables. To sum it up, the regression model exhibits how financial markets and macroeconomic fundamentals are intertwined and offers valuable insights into the relationship between S&P 500 and key economic indices. To conclude, the regression model highlights the close connection between financial markets and macroeconomic fundamentals and provides insightful information on the relationship between the S&P 500 and important economic indices

Code and Data Appendix:

The Stata dataset utilized in this analysis, along with the accompanying .do and .log files (in text format), which document all data handling, estimation procedures, and statistical/regression outputs, have been included as part of this submission.

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